

## 1.9 Rate Problems Solutions

*Notation: = joins values that are exactly equal;  $\approx$  introduces a rounded value. Step 1 applies the rate relationship, rate  $\times$  time = quantity or a rearrangement. Step 2 converts that result to the units asked for, carrying the unrounded Step 1 value. Convert a unit first when that makes Step 1 simpler; metric changes (m to km, cm to m) are done in the head.*

1. Sunlight.  $r \cdot t = d$ , so  $t = \frac{d}{r}$ .

The distance is in km, so first write the speed in km/s:  $3.00 \times 10^8 \text{ m/s} = 3.00 \times 10^5 \text{ km/s}$ .

Step 1:  $t = \frac{1.496 \times 10^8 \text{ km}}{3.00 \times 10^5 \text{ km/s}} = 498.666 \dots \text{ s}$

Step 2:  $498.666 \dots \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 8.3111 \dots \approx \mathbf{8.31 \text{ min}}$

*Three significant figures, set by the  $3.00 \times 10^8 \text{ m/s}$ . Rounding Step 1 to 499 s first gives 8.32 min, not 8.31: rounding too early changes the answer. The light we see left the Sun about eight minutes ago.*

2. Hair.  $t = 15 - 9 = 6 \text{ yr}$ .

Step 1:  $d = r \cdot t = 15 \frac{\text{cm}}{\text{yr}} \times 6 \text{ yr} = 90 \text{ cm}$

Step 2:  $90 \text{ cm} = \mathbf{0.90 \text{ m}}$

*Two significant figures, set by “about 15 cm”; the 6 years is a count and does not limit the answer. Keep the trailing zero: 0.9 m has only one significant figure.*

## 3. Pasta pot.

The rate is per minute, so first write the time in minutes:  $45 \text{ s} \times \frac{1 \text{ min}}{60 \text{ s}} = 0.75 \text{ min}$ .

$$\text{Step 1: } \text{quantity} = r \cdot t = 2.2 \frac{\text{gal}}{\text{min}} \times 0.75 \text{ min} = 1.65 \text{ gal}$$

$$\text{Step 2: } 1.65 \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 6.24525 \text{ L} \approx \mathbf{6.2 \text{ L}}$$

*Two significant figures: both 2.2 gal/min and 45 s have two. Accept a solution that converts the rate to gal/s instead (2.2 gal/min = 0.0366... gal/s); the answer is the same.*

4. Usain Bolt.  $r = \frac{d}{t}$ .

$$\text{Step 1: } r = \frac{100 \text{ m}}{9.58 \text{ s}} = 10.438 \dots \text{ m/s}$$

$$\text{Step 2: } 10.438 \dots \frac{\text{m}}{\text{s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 37.578 \dots \approx \mathbf{37.6 \text{ km/h}}$$

*Three significant figures, set by the 9.58 s; the 100 m is the exact race distance. This is his average speed — he starts from rest, and his top speed during the race was near 12.3 m/s, about 44 km/h.*